

# AHMED AKRAM AMER

AI / Machine Learning Engineer | AI Engineering, MLOps & LLM Applications

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## PROFESSIONAL SUMMARY

Electronic Engineering graduate in Computer Science and Engineering with an Excellent GPA of 3.71/4.0, focused on AI Engineering and MLOps. Hands-on experience building LLM applications with RAG, prompt engineering, and agent frameworks, alongside developing and deploying ML pipelines with Python, Docker, FastAPI, and CI/CD tools. Familiar with monitoring, workflow orchestration, and collaborative delivery of reliable AI systems.

## EDUCATION

### Menoufia University

B.Sc., Electronic Engineering (Computer Science & Engineering)

- GPA: 3.71 / 4.0 - Excellent

Sep 2021 - Oct 2026

Shebin el-Kom, Egypt

## WORK EXPERIENCE

### Tips Hindawi

AI Engineering Intern

- Internship - AI Engineering, Egypt (1 Month)
- Developed foundational knowledge of HuggingFace Transformers, transformer and attention-mechanism architectures, Groq, Streamlit, LangChain, and retrieval-augmented generation (RAG).
- Built and tested LLM applications, including a YouTube Video Summarizer and an Islamic QA assistant for the Egyptian dialect.

Jun 2026 - Jul 2026

Egypt

### SAiR

MLOps Intern

- Internship - MLOps Fundamentals and Best Practices
- Studied machine learning fundamentals, the ML lifecycle, and MLOps best practices for developing and maintaining reliable ML systems.
- Gained hands-on experience implementing MLOps workflows with FastAPI, Docker, Feast, Prefect, MLflow, GitHub Actions, Apache Airflow, Prometheus, and Grafana.

Jul 2026 - Sep 2026

### DEPI | EYouth

AI & Data Science Intern

- Internship - AI & Machine Learning, Egypt (Remote)
- Engineered supervised ML models for predictive analytics, achieving 90% accuracy on structured datasets.
- Reduced manual reporting time through automated ML pipelines deployed via Flask and Docker.
- Conducted EDA and feature engineering with Pandas and Seaborn across 10+ high-dimensional datasets.

Jun 2024 - Dec 2024

Egypt

## PROJECTS

### VisionAssist - AI-Powered Scene Description & Danger Detection [github.com/Ahmed-7-ML/RealTime\\_Scene\\_Description](https://github.com/Ahmed-7-ML/RealTime_Scene_Description)

- Python, FastAPI, Qwen2-VL-2B, Modal, Next.js, Spring Boot
- Built the Python and FastAPI AI captioning service for a real-time scene-description system that helps visually impaired users navigate via live camera feeds.
- Captioned frames and classified scenes as SAFE or DANGEROUS with a fine-tuned Qwen2-VL-2B model, deploying the service as a serverless GPU endpoint on Modal within a Next.js and Spring Boot pipeline.

### Islamic-QA-Egyptian-Dialect - Fine-Tuned Islamic Q&A Assistant [github.com/Ahmed-7-ML/Islamic-QA-Egyptian-Dialect](https://github.com/Ahmed-7-ML/Islamic-QA-Egyptian-Dialect)

- Python, Qwen2.5-7B-Instruct, LoRA/QLoRA, vLLM, LiveKit Agents
- Fine-tuned Qwen2.5-7B-Instruct on a curated Egyptian-Arabic Islamic Q&A dataset (Aqeedah, Fiqh, Seerah/history, and general questions) using LoRA/QLoRA.
- Deployed the merged 16-bit model via vLLM and integrated it into a real-time, voice-based AI conversational agent using the LiveKit Agents Framework, with a Deepgram STT → LLM → Cartesia TTS pipeline.

### QueryGuard AI - Enterprise Text-to-SQL Platform [github.com/Ahmed-7-ML/QueryGuard-AI](https://github.com/Ahmed-7-ML/QueryGuard-AI)

- Python, Gemini 2.5 Flash, Streamlit, SQLite3, RBAC
- Built a GenAI platform translating English/Arabic questions into optimized SQL queries for non-technical users.
- Implemented dual-layer RBAC: DROP, DELETE, and TRUNCATE blocked at both the UI and LLM prompt levels.

### Chat with PDF | RAG Application [github.com/Ahmed-7-ML/ReActify-PDF](https://github.com/Ahmed-7-ML/ReActify-PDF)

- Python, FastAPI, LangChain, Supabase (pgvector), HuggingFace, Groq
- Built an enterprise-grade, lightweight, high-performance PDF-chatting application powered by an agentic RAG core.

- Generated embeddings locally with HuggingFace sentence-transformers/all-MiniLM-L6-v2 and indexed them in a cloud-based Supabase Vector Database (PostgreSQL + pgvector).
- Enabled dynamic document querying using Groq's fast inference engine with the qwen/qwen3-32b model.

### Deep Learning - Multi-Domain

- PyTorch, TensorFlow, Scikit-learn, OpenCV
- Built end-to-end PyTorch pipelines for image classification (CIFAR-10, MNIST), audio classification, and breast cancer detection.

### ML Forge - End-to-End ML Lifecycle Platform

- Python, Flask, MLflow, Scikit-learn
- Architected ML Forge, a web platform automating the full end-to-end machine learning lifecycle, from data ingestion to deployment, via MLflow.

## EXTRACURRICULAR ACTIVITIES

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### NASA Space Apps Challenge

Oct 2024

#### Participant

- Seismic Detection Across the Solar System, Egypt
- Developed a seismic signal detection system for planetary data analysis.
- Presented the project in a final demo and discussion session, receiving a certificate of attendance.

## TECHNICAL SKILLS

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- **Programming Languages:** Python, Java, C++, C#
- **AI Agents:** LangChain, LangGraph, CrewAI, LangSmith, AgentOps, RAG, ReAct, Prompt Engineering
- **LLMs and NLP:** HuggingFace Transformers, Groq, Google Gemini, OpenAI, NLTK, spaCy
- **Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, OpenCV
- **Web and APIs:** FastAPI, Flask, Streamlit, Gradio
- **MLOps and Cloud:** MLflow, Docker, GitHub Actions (CI/CD), Feast, Prefect, Apache Airflow, Prometheus, Grafana
- **Databases:** SQLite, PostgreSQL, SQL Server, Oracle, Pinecone, Chroma, Qdrant
- **Tools:** Git, GitHub, VS Code, Jupyter, Google Colab
- **Soft Skills:** Problem Solving, Attention to Detail, Fast Learner, Adaptability, Communication, Teamwork & Collaboration

## CERTIFICATIONS

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- AWS Certified AI Practitioner: Amazon Web Services (AWS) - Aug. 9, 2026
- Building LLM Applications using Prompt Engineering: NVIDIA - Feb. 12, 2026
- Machine Learning Specialization: DeepLearning.AI - Aug. 29, 2024
- AI Engineer for Data Scientists Associate: DataCamp - Apr. 23, 2025
- HCIA-AI: Huawei - Sep. 3, 2024
- Python for AI and Data Science: IBM - Jul. 15, 2025
- Data Scientist: DEPI & EYouth - Dec. 2025
- Applied Deep Learning: MaharaTech - Aug. 2024
- AI Camp: Sprints × Microsoft

## LANGUAGES

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- Arabic(Native)
- English(Intermediate)